

Remote adversarial VR serious game simulating COVID-19 infection spread and protection protocols

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ABSTRACT

This paper presents Virtual Reality 3D interaction and simulation techniques in the context of the 2021 3DUI contest of the IEEE VR conference. This work discusses the design and development of a simulation based on COVID-19 indoor safety guidelines, aiming to familiarize and inform the potential users about the methods of transmission of SARS-CoV-2. These techniques are integrated in the context of a 2 players remote adversarial virtual reality serious-game, focusing on the education of the user, concerning airborne viruses transmission and health protection protocols through an entertaining gamified simulation.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

In the last year, humanity faced the challenge of COVID-19 pandemic, caused by coronavirus SARS-CoV 2, resulting to unprecedented global protection measures. This epidemiological situation stimulated, not only the medical community with research on vaccination, treatment and transmission, but also data scientists with data science techniques on detection, mitigation, treatment and elimination [1] of COVID-19.

Our proposed 3D user interface design, aims to inform the end-users about the coronavirus infection spread and protection protocols, through a virtual reality remote adversarial serious game. The aforementioned interactive user interface presents a gamified scenario, focusing on health protection protocols, simulating airborne virus transmission. Section 2 highlights the simulation techniques used, following adequate literature. Section 3 outlines the core concepts and gamification elements of our serious-game design. Section 4 discusses the challenges faced and possible extension of the present work.

2 SIMULATION DESIGN

In this section we will examine the design of the proposed system, regarding the latest research in airborne transmitted diseases. The application developed, simulates the potential of the virus transmission in an indoor environment. Recent studies suggest three primary directions of human-to-human transmission of COVID-19 [2]. All three are based in viral droplet transfer to the recipient's respiratory mucosae. This can be achieved either by inhalation of large drops and microdroplets ejected by an infected person or by physical contact with droplets from the hands of the recipient to their mouth, nose or eyes.

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The so-called 'airborne' transmission refers to the inhalation of the aerosol microdroplets, remaining suspended for a significant amount of time within indoor spaces. A modified Wells-Riley equation proposed by Rudnick and Milton [3] was utilized for the calculation of the probability of infection from these droplets in the enclosed environment of the supermarket environment. The equation can be described as follows:

$$P = 1 - \exp \left\{ -\frac{iqpt}{Q} \times \left[1 - \frac{V}{Qt} \left(1 - \exp \left(-\frac{Qt}{V} \right) \right) \right] \right\} \quad (1)$$

where P is the risk of infection for susceptible population, i is the number of infectors, q is the quantum generation rate by an infected person (quanta/h), p is the breathing rate per person (m^3/h), t is the total exposure time (hours), Q is the outdoor air supply rate (m^3/h), and V is the volume of the ventilated space (m^3). Regarding SARS-CoV-2, the quantum generation rate is $q = 142$ quanta/h, as proposed by an estimation of airborne viral emission [4] and we estimate the breathing rate $p = 0.5 m^3/h$, per healthy and $p = 0.75 m^3/h$ per infected person, as suggested by the same and another model [4, 5]. The air supply rate Q is calculated by the store conditions (ventilation, humidity, breathing exchange rate) and the volume of the space V is calculated by the store dimensions.

While aerosol droplets play a dominant role in the spread of COVID-19, the risk of infection from larger droplets, applied in close range situations ($<2m$), is relatively bigger. In close range situations, the original Wells-Riley equation [6] is used:

$$P = 1 - \exp \left(-\frac{iqpt}{Q} \right) \quad (2)$$

Large droplets, being produced at close distance between people (i.e. by coughing or sneezing), have about 6 times the concentration of the viral pathogen than aerosol particles [7]. Respectively, the quantum generation rate can be estimated as $q = 6 \times 142 = 852$ quanta/h for close distance probability calculations.

It is well established that when face masks are worn by the infected and the susceptible individuals, the risk can be reduced. In this model, we demonstrate the impact of face masks with a 60% reduction in the quanta generation rate q , when worn by an infected individual and by increasing the airflow rate Q by 60% when worn by a susceptible person, as proposed in [8].

Finally, an instant infection from physical contact is taken in consideration. The above probabilities for airborne infection are calculated every 1 second. When one susceptible individual considered infected, for game purposes, we make the assumption that they begin to spread the virus instantly.

In conclusion, we highlight the primary routes of COVID-19 transmission and calculate the infection risk for all three. Consequently, we present a realistic simulation of the COVID-19 infection spread in an indoor environment, valuable for guidance and information.

3 GAME DESIGN

The game was designed in Unity3D engine utilizing the Unity XR plug-in framework in order to support cross platform experiences.

For the remote syncing of the networked components we made use of the free version of Normcore, a multiplayer package supporting XR compatibility and networked physics. Through the development process, the Oculus Rift and Oculus Quest 2 head-mount displays were used. The game takes place in a small town supermarket during



Figure 1: The supermarket with the NPCs

the COVID-19 pandemic. The first player (A) represents the health manager of the supermarket, aiming to protect the customers by keeping the facility clean and safe, while the second player (B), being already infected by the virus, is intent upon spreading the virus. All the customers of the game are Non Player Characters (NPCs) spawned randomly from the game AI.

Given that player B is infected their goal is to infect as many surfaces and customers as they can. This can be achieved either by transmitting the virus to a customer at once or by infecting a surface with their hands and transmitting it implicitly. While player B embraces the characteristics of the airborne viruses transmission inside indoor spaces, player A has several tools, spawned gradually, in order to achieve a game equilibrium. The tools used are :

- A pistol, shooting mask projectiles, used to supply the customers with masks.
- An eye-wear with special lenses, in order to identify the infected areas.
- A cleaning spray, for the sterilization of the aforementioned contaminated areas.

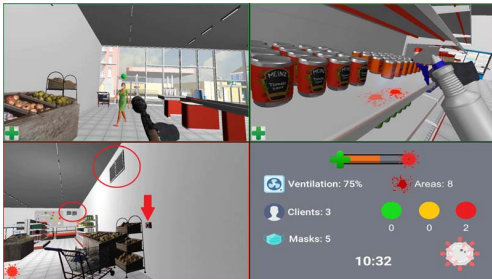


Figure 2: The mask gun, the cleaning spray, the supermarket's ventilation and the UI of the game

NPCs enter the supermarket space, and navigate, following randomly chosen paths determined by target positions with unspecified wait time in each target. Each of them enters the supermarket in a healthy state, and can get infected:

- while browsing a contaminated area or product
- by crowding in spaces with other infected NPCs
- when getting into close contact, for a short time interval, with player B or player A if they are infected.

Apart from the aforementioned features and mechanics, both players can follow various strategies, in order to outplay their opponent. Each of them, using their personal weapons, can get a time leverage over their enemy or use other objects of the environment artfully. Player A can freeze their opponents movement for some seconds by using the cleaning spray, while player B can infect and keep them on quarantine, respectively. Additionally, several objects inside the environment are available to both players, like ventilation control mechanisms and objects used to maintain/avoid crowding or block the opponent's movement.

Several indication elements, such as visualizations of aerosol droplets and infection probabilities on the NPCs, while some gamification elements such as in-game challenges (i.e. throw 10 masks, increase air ventilation rate etc) help the user get familiar with airborne viruses transmission. The game has a point system depending on the level of the virus spread in the store, calculated by the model described in section 2 and the number of the infected customers. The air contamination can be controlled through the venting system which is available to both players to activate and deactivate. Time is scaled properly for game purposes. The players can see the clock counting down and have to hurry to achieve their goal.

The minimum system requirements are presented in Table 1. Each user must be equipped with a VR HDM and have a broadband Internet connection.

CPU	Intel i3-3100/AMD Ryzen 3 1200, FX4350
GPU	NVIDIA GTX 1050Ti/AMD Radeon RX 740
RAM	8GB

Table 1: Minimum System Specifications

4 CONCLUSION

The present work discusses the design and development of an instructional remote multiplayer VR game, focusing on the education of the users not only about the COVID-19 situation, but airborne viruses transmission and health protocols in general. Apart from the gamification elements, an airborne virus transmission simulation was developed, based on the literature, in order to emulate realistic conditions in the serious-game scenario.

REFERENCES

- [1] S. Latif, M. Usman, S. Manzoor, W. Iqbal, J. Qadir, G. Tyson, I. Castro, A. Razi, M. N. K. Boulos, A. Weller, and J. Crowcroft. Leveraging data science to combat covid-19: A comprehensive review. *IEEE Transactions on Artificial Intelligence*, 1(1):85–103, 2020.
- [2] Rajat Mittal, Rui Ni, and Jung-Hee Seo. The flow physics of covid-19. *Journal of Fluid Mechanics*, 894:F2, 2020.
- [3] SN Rudnick and DK Milton. Risk of indoor airborne infection transmission estimated from carbon dioxide concentration. *Indoor air*, 13(3):237–245, 2003.
- [4] Giorgio Buonanno, Luca Stabile, and Lidia Morawska. Estimation of airborne viral emission: quanta emission rate of sars-cov-2 for infection risk assessment. *Environment International*, page 105794, 2020.
- [5] Martin Z Bazant and John WM Bush. Beyond six feet: A guideline to limit indoor airborne transmission of covid-19. *medRxiv*, 2020.
- [6] EC Riley, G Murphy, and RL Riley. Airborne spread of measles in a suburban elementary school. *American journal of epidemiology*, 107(5):421–432, 1978.
- [7] LJGR Morawska, GR Johnson, ZD Ristovski, Megan Hargreaves, K Mengersen, Steve Corbett, Christopher Yu Hang Chao, Yuguo Li, and David Katoshevski. Size distribution and sites of origin of droplets expelled from the human respiratory tract during expiratory activities. *Journal of Aerosol Science*, 40(3):256–269, 2009.
- [8] Amelia Harrichandra, A Michael Ierardi, and Brian Pavilonis. An estimation of airborne sars-cov-2 infection transmission risk in new york city nail salons. *Toxicology and industrial health*, 36(9):634–643, 2020.